

Exercise class 8

Inhomogeneous and higher-order linear ODE's

1. *RLC circuits, solution of switching off protocol:*

$$\ddot{U} + \frac{R}{L}\dot{U} + \frac{1}{LC}U = 0, \quad U(0) = U_0, \quad \dot{U}(0) = 0$$

2. *Linear (in)dependence:* Verify that the functions

$$y_1 = \cos(3 \arccos x), \quad y_2 = 6x - 8x^3$$

are particular solutions of the differential equation

$$x^2 y'' - 3xy' + 3y = 0.$$

Check whether they are linearly independent by calculating the Wronskian

$$W(x) = y_1 y_2' - y_2 y_1'.$$

Evaluate $W(x)$ for a few x . [Suggestions: $x = 0, x = 1/2$.]

3. *2nd order inhomogeneous linear ODE (Kreyszig 2.7)*
2.7 Problem 12, 14, 18

4. *Bending Beam (Kreyszig 3.3)*
Find the general solution and solve the initial value problem.

$$\frac{d^4 y}{dx^4} = k$$

$$y(0) = y'(0) = 0, \quad y(L) = y'(L) = 0$$

5. *Higher order ODE's*
3.2 Problems 1, 6
3.3 Problem 1